

HR Analytics Dashboard Project Report

1. Project Overview

This project analyzes employee workforce data using Python, SQL, and Power BI to understand employee demographics, salary distribution, job satisfaction, performance trends, and employee attrition patterns. The objective is to identify factors influencing employee retention and provide data-driven recommendations that support workforce planning and organizational growth.

2. Dataset Summary

The HR Analytics dataset contains employee-related information collected from various departments within an organization.

Key Features

- Employee Demographics (Age, Gender, Marital Status, Education Field)
- Organizational Information (Department, Job Role, Job Level)
- Compensation Metrics (Monthly Income, Salary Hike)
- Performance Metrics (Performance Rating, Job Satisfaction)
- Employment Details (Years at Company, Total Working Years, Overtime)
- Attrition Information (Employee Resignation Status)

Business Objective

The primary objective of this analysis is to:

- Identify employee attrition patterns.
- Understand salary distribution across different roles.
- Analyze employee satisfaction and performance.
- Evaluate promotion and career growth opportunities.
- Support HR decision-making through data-driven insights.

3. Exploratory Data Analysis using Python

We began with data preparation, cleaning, and exploratory analysis using Python.

Data Loading

- Imported the HR Analytics dataset using Pandas.
- Loaded data into DataFrames for analysis and preprocessing.

Initial Exploration

- Examined dataset structure using:
 - `df.info()`
 - `df.describe()`
 - `df.head()`
- Verified data types and overall dataset quality.

Data Cleaning

- Checked for missing values.
- Identified duplicate records.
- Ensured consistency across employee-related attributes.

Exploratory Data Analysis

The following analyses were performed:

- Employee distribution by gender.
- Department-wise employee count.
- Attrition distribution across departments.
- Salary distribution across job roles.
- Job satisfaction analysis.
- Performance rating evaluation.
- Overtime and employee retention analysis.

Data Visualization

Python visualizations were used to identify trends and support decision-making regarding:

- Employee attrition.
- Workforce demographics.
- Salary structure.
- Employee performance.

Dashboard Preparation

Processed and transformed data was prepared for SQL analysis and Power BI dashboard development

4. Data Analysis using SQL (HR Analytics)

We performed structured analysis in SQL to answer key workforce and employee retention questions.

1. Total Monthly Income by Gender

Compared total monthly income earned by male and female employees to understand compensation distribution.

	Gender	total_income
▶	Male	5471389
	Female	3808771

2. Attrition Count by Department

Identified departments experiencing the highest employee attrition.

Department	attrition_count
Research & Development	127
Sales	91
Human Resources	12

3. Average Monthly Income by Job Role

Analyzed salary differences across job roles.

JobRole	avg_income
Manager	17144.90
Research Director	16078.04
Healthcare Representative	7512.17
Manufacturing Director	7309.04
Sales Executive	6936.98
Human Resources	4194.35
Laboratory Technician	3248.15
Research Scientist	3234.85
Sales Representative	2630.29

4. Education Field Attrition Analysis

Determined which educational backgrounds show the highest attrition levels.

EducationField	attrition_count
Life Sciences	88
Medical	57
Marketing	35
Technical Degree	32
Other	11
Human Resources	7

5. Job Satisfaction by Role

Evaluated average job satisfaction scores across different job positions.

jobRole	avg_satisfaction
Human Resources	2.53
Manufacturing Director	2.65
Laboratory Technician	2.71
Sales Representative	2.72
Manager	2.72
Research Director	2.73
Sales Executive	2.74
Healthcare Representative	2.76
Research Scientist	2.79

6. Experienced High Performers

Identified employees with more than 10 years of experience who maintain excellent performance ratings.

EmpId	Age	JobRole	TotalWorkingYears	PerformanceRating
RM809	29	Research Scientist	11	4
RM045	30	Laboratory Technician	12	4
RM212	30	Manufacturing Director	12	4
RM411	30	Manufacturing Director	11	4
RM603	30	Manufacturing Director	12	4
RM887	30	Research Scientist	12	4
RM435	31	Manufacturing Director	13	4
RM485	31	Sales Executive	13	4
RM1406	31	Research Director	13	4
RM237	33	Laboratory Technician	13	4
RM510	33	Healthcare Represent...	15	4

7. Department-wise Employee Income Ranking

Ranked employees within departments based on monthly income using SQL window functions.

EmpId	Department	MonthlyIncome	income_rank
RM957	Human Resources	19717	1
RM1155	Human Resources	19658	2
RM1402	Human Resources	19636	3
RM539	Human Resources	19189	4
RM536	Human Resources	19141	5
RM106	Human Resources	18844	6
RM478	Human Resources	18200	7
RM113	Human Resources	17328	8
RM1000	Human Resources	16799	9
RM1097	Human Resources	16437	10
RM963	Human Resources	14026	11

8. Top 3 Highest Paid Employees per Department

Retrieved top-paid employees from each department for compensation benchmarking.

EmpId	Department	MonthlyIncome	rnk
RM957	Human Resources	19717	1
RM1155	Human Resources	19658	2
RM1402	Human Resources	19636	3
RM191	Research & Development	19999	1
RM747	Research & Development	19973	2
RM852	Research & Development	19943	3
RM919	Sales	19847	1
RM750	Sales	19845	2
RM1243	Sales	19833	3

9. High Performers with Below-Average Salary

Identified employees whose performance is high but compensation remains below organizational averages.

EmpId	JobRole	MonthlyIncome	PerformanceRating
RM172	Sales Representative	2325	4
RM178	Laboratory Technician	1102	4
RM854	Research Scientist	2552	4
RM893	Research Scientist	1859	4
RM916	Laboratory Technician	2625	4
RM161	Research Scientist	2323	4
RM499	Research Scientist	2773	4
RM631	Manufacturing Director	4775	4
RM1340	Research Scientist	2472	4
RM586	Laboratory Technician	1601	4
RM911	Research Scientist	1223	4

10. Years at Company vs Attrition Trend

Analyzed the relationship between employee tenure and attrition.

YearsAtCompany	total_employees	attrition_count
0	42	16
1	162	57
2	123	26
3	120	21
4	107	18
5	193	20
6	75	9
7	90	11
8	78	9
9	81	8
10	115	16

11. Gender-wise Promotion Potential

Compared average job levels between male and female employees.

Gender	avg_job_level
Male	2.04
Female	2.12

12. Overtime Impact on Income

Examined how overtime work influences employee income levels.

OverTime	avg_income
No	6490.65
Yes	6598.67

5. Dashboard Development in Power BI

An interactive HR Analytics Dashboard was developed in Power BI to visualize key workforce metrics.

Dashboard Components

- Employee Count KPI
- Attrition Rate KPI
- Department Distribution
- Gender Distribution
- Monthly Income Analysis
- Job Satisfaction Analysis
- Attrition by Department
- Overtime Impact Analysis
- Performance Rating Insights

- ❖ We built an interactive dashboard in **Power Bi** to present insights visually.



6. Key Insights

Workforce Distribution

Research & Development contains the largest workforce, followed by Sales and Human Resources.

Attrition Trends

Employee attrition is concentrated within specific departments and job roles, indicating potential retention challenges.

Compensation Analysis

Significant salary variation exists across job roles and organizational levels.

Job Satisfaction

Certain job roles display lower satisfaction levels, suggesting areas requiring management attention.

Overtime Impact

Employees working overtime demonstrate different compensation and retention patterns compared to non-overtime employees.

Performance Evaluation

Several high-performing employees are compensated below organizational averages, indicating potential retention risks.

7. Recommendations

Improve Employee Retention

Develop targeted retention strategies for departments and job roles experiencing high attrition.

Enhance Employee Engagement

Implement programs focused on employee satisfaction, recognition, and career development.

Compensation Review

Review salary structures to ensure fair compensation for high-performing employees.

Promotion Planning

Create transparent career growth opportunities to improve employee motivation and retention.

Overtime Management

Monitor overtime workloads to reduce burnout and improve work-life balance.

Predictive HR Analytics

Adopt predictive models to identify employees at risk of leaving the organization.

8. Conclusion

This HR Analytics project demonstrates how Python, SQL, and Power BI can be integrated to transform raw employee data into actionable business insights. The analysis highlights key workforce trends, attrition patterns, compensation structures, and employee satisfaction indicators. By implementing the recommendations derived from this study, organizations can improve retention, optimize workforce planning, and enhance overall employee performance.